PROCESS AND INTEGRATION DOCUMENT

Workflow and Integration Diagrams

How a vehicle sighting travels from a camera to an officer, and how Sentinel connects to the systems around it

PROJECT	Sentinel
COVERS	Sighting lifecycle, matching, integration, onboarding, recovery
COMPANION TO	High Level Design and Architecture
DOCUMENT	2 of 3, Workflow and Integration

The life of a vehicle sighting

Six stages, from photons at a camera to a path drawn on an officer's map. Stages one to four happen at the edge and never leave the district. Only the record produced at stage four travels.

1 Decode the sub stream

PyAV opens the camera over RTSP and decodes the sub stream, not the main stream. Timestamps are taken from the video container, so timing is when the frame was captured and not when the network delivered it.

2 Detect vehicles

ONNX Runtime runs a YOLO or RT DETR export on the frame and returns boxes with a class and a confidence. GPU when present, CPU when not.

3 Track across frames

Each vehicle is followed between frames and given a local track identity. The tracker measures time in seconds taken from the video, so a dropped connection cannot corrupt the timeline.

4 Read the plate and describe the vehicle

The plate is located on the vehicle crop rather than the whole frame, characters are read, then corrected against Indian plate grammar. In parallel a 512 dimension appearance vector, plus colour and vehicle type, is produced.

5 Emit one small record

About 400 bytes: camera, time, plate and its confidence, vehicle type, colour, direction, appearance vector. This is the only thing that leaves the site. Video stays in the local ring buffer.

6 Match, alert and display

The state core applies the road network gate, fuses the evidence, writes the sighting and pushes it to the command centre over WebSocket. Watchlist and restricted zone rules fire here.

STAGES 1 TO 4 RUN ON THE EDGE NODE. STAGE 5 IS THE ONLY THING THAT CROSSES THE NETWORK.

Cross camera matching and the road network gate

This is the step that makes cross camera tracking trustworthy rather than merely fast. Geography removes almost every candidate before any comparison is made, so the matcher is only ever asked an easy question.

20,000
All vehicles in the window

↓

1,900
Reachable by road in the elapsed time

↓

473
Correct direction of travel

↓

1 to 4
Survives appearance and plate fusion

97.6%	2,240 ms	53 ms	42x
Fewer comparisons	Before the gate	After the gate	Faster end to end

WHY THE GATE EXISTS

The reason this matters is accuracy, not speed

A matcher that is 99 percent accurate sounds excellent. Run it against 20,000 candidates and the one percent becomes roughly 200 wrong links every hour, which no officer can audit and which quietly destroys trust in the system. Run the identical matcher against 473 candidates and the same error rate produces about four, which a person can actually check. The gate does not make the model better. It makes the question small enough for the model to be right.

SAFETY RULE

The ceiling that cannot be argued with

After fusion, a link supported only by appearance evidence is labelled probable and can never be labelled confirmed. Confirmed requires a plate read. This is enforced in the matcher itself rather than left to an operator guideline, because a system that can present a colour and shape guess as a positive identification will eventually accuse the wrong person.

EXAMPLE

A worked example

A white hatchback is seen at Naroda junction at 14:23:45. Three minutes later the question is not which of twenty thousand vehicles this might be. It is which of the four junctions reachable from Naroda by road in three minutes, facing the right way, saw a white hatchback. That is a question with a checkable answer, and it is the same question a good investigating officer would ask.

What Sentinel connects to

Sources on the left, Sentinel in the middle, consumers and external systems on the right. Solid borders are built and tested. Dashed borders await an external approval that is not a software task.

CAMERA SOURCES

RTSP
IP cameras and DVR channels, main and sub stream

HLS
Municipal, toll and smart city feeds, and the RTSP fallback

WHEP
WebRTC playback for live view in a browser

ONVIF discovery
Auto onboarding for roughly half the estate

Sentinel catalogue
Gateway supplied camera list, reconciled on start

SENTINEL

Ingestion service
Decode, detect, track, read, embed

Event processor
Gate, fuse, persist, alert

PostgreSQL 16
Sightings, cameras, users, audit

Backend API
42 REST endpoints and a WebSocket

CONSUMERS AND EXTERNAL SYSTEMS

Command centre
Map, search, alerts, camera health

Officer mobile browser
Same API, same isolation rules

Prometheus and Grafana
Metrics, dashboards, alerting

VAHAN and SARTHI
Registration and licence lookup

eGujCop
Case and FIR linkage

AFIS and NAFIS
Fingerprint systems

STATUS

On the five government systems, be precise

The adapters are written, typed and tested against mock backends, and each one enforces a stated purpose and writes an audit record before any request leaves. What they are not is connected to a live endpoint, because each system belongs to a different authority and each needs its own approval. That is an administrative dependency and not an engineering gap, and it should be stated rather than discovered.

DESIGN DETAIL

One protocol note that decides whether a deployment succeeds

The pipeline consumes the camera sub stream rather than the main stream. Almost every IP camera and DVR channel of the last decade publishes both: a main stream at 1080p or 4K, and a sub stream at D1 or 720p. Detection, colour, type and appearance matching all work on the sub stream, and using it cuts decode and inference cost by six to eight times. The main stream is recorded locally and pulled only when a human needs to read a plate or a face at full resolution. This single choice is the difference between three GPUs and twenty five for the same camera count.

Onboarding a camera, and surviving a blocked port

Adding cameras is a configuration task, not a development task. No code changes, no rebuild, no downtime for the rest of the estate.

1 Collect three facts
Stream address, credentials, and position with a compass heading

2 Add one entry
One block in config cameras yaml, credentials held separately

3 Restart one service
Only the ingestion service, the rest of the system keeps running

4 Probe and go live
The camera is probed once for codec, resolution and frame rate, then marked

SURVEY REQUIREMENT

The field detail that is most often missed

Every camera needs a compass heading. Without it the system knows where a camera is but not what it is looking at, and the road network gate is directional, so a missing heading materially weakens the strongest part of the design. Thirty seconds per camera during installation. Very expensive to retrofit later.

Transport failover, because government networks block RTSP

Try RTSP over TCP
Port 8554. The normal path and the one the AI pipeline prefers.

Count consecutive failures
A single blip is not a failover. Repeated failure is.

Rotate to HLS over 443
Same camera, different wire. The reader keeps cycling rather than giving up.

Why this is worth building before the first deployment

Many government networks close port 8554. On such a network, without failover, every camera reports as offline and the operator sees a wall of red. The natural conclusion is that the system has failed, when in fact the network is doing exactly what it was configured to do. Automatic rotation turns a false emergency into a line in a log.

CORRECTNESS

What a stream break does to the tracker

A reconnect is not a continuation. When a stream drops and resumes, the scene may have moved on by seconds or minutes, so carrying the old track identities forward would invent journeys that never happened. The reader detects the discontinuity, resets the tracker and clears the appearance gallery, and the timeline restarts from the container timestamps rather than from the moment the socket reopened.

Access, audit and recovery

Every request from an officer passes the same four checks before a row is returned. None of them can be skipped by a developer who forgets, because they are not in application code.

1 Authenticate
Valid token, active account, not locked out

2 Authorise the role
One of six roles, least privilege by default

3 Scope to department
Written by a database trigger, indexed, 0.72 ms

4 Record the purpose
Stated reason stored for seven years

AUDIT

Reading the audit trail is itself audited

An auditor role exists that can read the audit log and nothing else. When it does, that read is written to the audit log. The question an investigator is eventually asked is not only who looked up this citizen, but who went looking to find out. Both are answerable.

Recovery, measured rather than described

A police evidence system that loses the last few minutes of sightings on a power failure has lost evidence, not throughput. So it was tested rather than documented. The drill builds a primary and a streaming standby, writes 5,000 evidence rows, kills the primary with no clean shutdown, promotes the standby and counts what survived.

0	0.36 s	5,000	Yes
Records lost	To accept writes again	Rows written before the kill	WAL archived for point in time recovery

Streaming replication and archiving answer two different failures

Replication protects against losing a machine, which is the failure everyone plans for. WAL archiving protects against a bad write that nobody notices for an hour, which is the failure that actually destroys evidence, because replication faithfully copies the damage to the standby. Both are configured.

OPERATIONS

The lockout, and the way back in

Five failed logins freeze an account for fifteen minutes, which stops credential stuffing without letting anyone lock a real officer out of a live incident indefinitely. Because a lockout is useless without an operator side, there is an audited recovery tool: it shows which accounts are locked and for how long, clears a lock, and resets a password using the same hashing the login path verifies. Every use writes an audit row, and the password itself is never written to it.